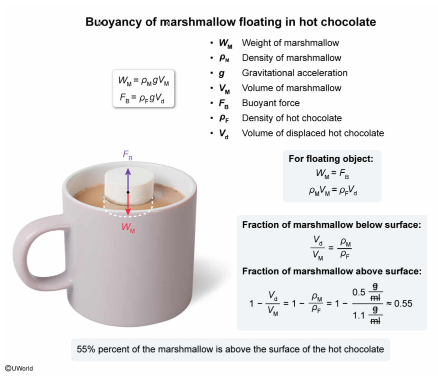
A 20-cm^3 marshmallow is placed in 350 cm^3 of hot chocolate. The densities of the marshmallow and hot chocolate are 0.5 g/ml and 1.1 g/ml, respectively. Ignoring the effects of atmospheric pressure, what percentage of the marshmallow is above the surface of the hot chocolate? (Note: Gravitational acceleration equals 10 m/s^2.)

- A. 6% [5%]
- B. 17% [33%]
- ✔ C. 55% [57%]
- D. 100% [3%]

Correct
57% answered correctly

Explanation:



The weight *W* of an object equals the product of its **density** *ρ*, gravitational acceleration *g*, and volume *V*:

$$W = \rho g V$$

Archimedes principle states that an object placed in a fluid experiences a buoyant force F_B equal to the product of the density ρ_F of the fluid, *g*, and the volume V_d of the displaced fluid:

$$F_B = \rho_F g V_d$$

When the object floats on the surface of the fluid, the value of F_B equals *W* of the object. Hence, it follows that:

$$F_B = W$$

$$\rho_F g V_d = \rho g V$$

$$\rho_F V_d = \rho V$$

In this question, the marshmallow has density $\rho = \rho_M$ and volume $V = V_M$. Solving the above equation for V_d yields:

$$V_d = \frac{\rho_M V_M}{\rho_F}$$

V_d represents the displaced volume of hot chocolate, which equals the submerged volume of the marshmallow. The volume V_{MA} of the marshmallow above the surface of the hot chocolate is the difference between V_M and V_d:

$$V_{MA} = V_M - V_d$$

$$V_{MA} = V_M - \frac{\rho_M V_M}{\rho_F}$$

Furthermore, the fraction of the marshmallow above the surface is the ratio of V_{MA} and V_M:

$$\frac{V_{MA}}{V_M} = \frac{\cancel{V_M}}{\cancel{V_M}} - \frac{\rho_M \cancel{V_M}}{\rho_F \cancel{V_M}}$$

$$\frac{V_{MA}}{V_M} = 1 - \frac{\rho_M}{\rho_F}$$

Substituting the values for ρ_M and ρ_F into this equation gives:

$$\frac{V_{MA}}{V_M} = 1 - \frac{0.5\ \frac{g}{\cancel{ml}}}{1.1\ \frac{g}{\cancel{ml}}} \approx 1 - 0.45 = 0.55$$

The percentage of the marshmallow above the surface equals 100 times this ratio. Therefore, 55% of the marshmallow is above the surface of the hot chocolate.

(Choice A) 6% could be calculated if the volumes of the marshmallow and hot chocolate are used in the equation for V_d rather than their densities.

(Choice B) 17% is obtained by using the difference between the densities of the hot chocolate and the marshmallow in the denominator of the final equation instead of the hot chocolate density.

(Choice D) 100% could be calculated if the densities of the marshmallow and hot chocolate are switched in the equation for V_d.

Educational objective:
An object floating in a fluid experiences a buoyant force that is equal to the weight of the displaced fluid. The proportion of the object that is submerged depends on the ratio of the densities of the object and the fluid.

Time spent:0 sec
Subject:
Physics

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Fluids